



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Information Technology
In-Semester Examination I
Academic Year: 2025-26 (Semester - I)

Class: SY B.Tech	Subject: Object Oriented Programming	Subject Code: PCC-IT202/203
Date: 22.08.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.
- iv. Figures to the right indicate full marks.
- v. The symbols carry the usual meaning.
- vi. For Q. No. 2.a and 2.b answer without proper explanation will get 0 marks.

Q.N.	Question	Marks	CO	BT
Q.1	a Explain benefits of object-oriented programming	5	CO1	BT1
	b Explain following concepts in brief in the context of Object-oriented programming: 1. Class and Object 2. Inheritance 3. Polymorphism 4. Encapsulation 5. Abstraction	5	CO3	BT2
Q.2	a Find the output of following program <pre>#include<iostream> using namespace std; int main() { int a[] = { 2, 4, 6, 8, 10 }; int i, sum = 0, * b = a + 4, x = 4; for (i = 0; i < 5; i++, x--) { sum = sum + (*b - i) + x; cout << sum; cout << "\n"; } return 0; }</pre>	5	CO2	BT5
	b Find the output of following program <pre>#include<iostream> using namespace std; int func2(int z) { if (z == 1) return 1; return (z * func2(z - 1)); } void func1(int* y) { int i; if (*y == 0)</pre>	5	CO2	BT5

		<pre> return; int x = *y; for (i = x; i >= 1; i--) { int result = func2(i) +x; cout << result; cout << "\n"; } } int main() { int x = 5; func1(&x); return 0; } </pre>			
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Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Be able to understand the basics of OOP and Object-oriented approach
CO 2	Be able to test, document and prepare a professional looking software
CO 3	To be able to apply an object-oriented approach to programming and identify potential benefits of object-oriented programming over other approaches
CO 4	To read from and write to files using file streams, handle text and binary files, and perform operations such as reading, writing, and appending data.
CO5	To be able to design software to solve real world problems based on object-oriented principles



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

ISE-II Examination (Semester-I) 2024-25

SY (CIVL/ CSE/ EXTC/ INST/ PROD/ CHEM/ TEXT/ ELEC/ MECH)

Python Programming

OE-IT201

Date: 24-10-2024

Time: 14:30 to 15:15

Max. Marks: 20

Instructions: 1. Make suitable assumptions wherever necessary. 2. All Questions are compulsory.

Q.N		Marks	CO OE-IT 201.X	BT X
Q. 1	Carefully review the following Python instructions and give answers based on how they would have resulted in a python interpreter. Provide a valid justification for your answers, without it no marks will be awarded.	[6]	1-3	1-5
	a. <pre>>>> a = [3,4] >>> a.extend("sggs") >>> print(a)</pre>			
	b. <pre>>>> b = {1:20, 2:30} >>> b.get('1')*20</pre>			
	c. <pre>>>> c = tuple(["sggs", 2, 0, 2, 4]) >>> c.index(2, 2)</pre>			
Q. 2	Construct a dictionary that stores information about three students, including their names, ages, and marks of any five subjects. Write a function to find and print the average marks of each student.	[7]	1-3	2-6
Q. 3	Implement a function called <i>calculate_area</i> that takes three parameters: length, width, and radius. This function should return the area of a rectangle and the area of a circle. In your program, avoid printing anything within the <i>calculate_area</i> function itself. print the result outside function. Finally provide the description about your logic in 3 to 4 sentences.	[7]	1-3	2-6

"Python was created to be fun to use, and it has stayed that way throughout its evolution."- Guido Van Rossum
Creator of Python Programming language. All The BEST!

Course Outcomes (CO)

OE-IT201.1 Utilize Python basics through the REPL environment, demonstrating proficiency in executing code snippets and exploring language features interactively.

OE-IT201.2 Implement Python Programs using core data structures like Lists, Dictionaries and use Regular Expressions.

OE-IT 201.3 Able to understand and write python programming using conditions, loops, functions etc.

OE-IT201.4 Able to understand and write python programming using modules, packages, strings etc with error handling.

OE-IT201.5 Able to understand and write python programming using files with error handling.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



Class: SY.

Course: Data Structures

Course Code: PCC-IT201

Date: 21.08.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Write proper Formula/ Theorem / Equation if question based on that.
- Write proper justification for your answer and solve every question step by step.

Q.N		Marks	CO	BT
	Write a C program to create an array of size n with condition 'n' must be ≥ 8 and 'n' can be odd also. After array creation insert any number at exactly the middle of the array and combine the array in sorted reverse order. (e.g. Input array can be $a[10, 5, 30, 20, 40, 60, 50, 65]$ after inserting 100 at middle the final array become $a[30, 20, 10, 5, 100, 65, 60, 50, 40]$)	5	CO2	BT1, BT2, BT3
	OR			
Q.1	Find the location of element in the given array with row-major order. Write the standard formula for same. a. Loc (A [375]) if array index as A [-250...575], base address is 8050 and size of each element is 14 bytes. b. Loc (B [175][35]) if array index as B [-500... -200, -185 ... 75], base address is 12500 and size of each element is 5 bytes.	5 [2,3]	CO2	BT1, BT2, BT3
Q.2	You are given a sorted array $a[]$ of n distinct integers. Your task is to write a C program that find any index i such that $a[i] = i$. If such an index exists, the program should output the index and the corresponding value. If no such index exists, the program should indicate that no such element is found. For searching use binary search algorithm only and find what is the complexity of program.	5	CO1	BT2, BT3
Q.3	You are given an infix expression that contains integers and standard arithmetic operators. Your task is to evaluate the postfix expression for the given infix expression : $5 * 6 - 8 + 8 / 2 \wedge 3 \wedge 1 * 4 - 7$. Solve step by step using stack with proper table format.	5[3,2]	CO2	BT3, BT4
Q.4	Hofstadter's Q-Sequence is a well-known example of a recursive sequence generated by the recurrence equation: $Q(1) = 1$; $Q(2) = 1$; and $Q(n) = Q(n - Q(n - 1)) + Q(n - Q(n - 2))$ for $n > 2$. Calculate $Q(5)$ and analyse its output tree step by step for all the recursive calls. Also Find the highest level of stack and total number of function calls.	5[3,2]	CO2	BT4

"Bad programmers worry about the code. Good programmers worry about data structures and their relationships"

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understanding the concept of Dynamic memory management, data types, algorithms, Big O notation.
CO 2	Understand abstract data types, basic data structures such as arrays, linked lists, stacks, and queues.
CO 3	Understand non-linear data structures.
CO 4	Describe the hash function and concepts of collision and its resolution methods.
CO 5	Apply Algorithm for solving problems like sorting, searching, insertion and deletion of data.